In aqueous solutions, when complex ion dissociates, there will be equilibrium between undissociated complex ion and dissociated ions. Hence the stability of the metal complex can be expressed in terms of dissociation equilibrium constant or instability constant (α). For example let us consider the dissociation of $\left[Cu(NH_3)_4\right]^{2+}$ in aqueous solution.

$$\left[Cu(NH_3)_4\right]^{2+} \rightleftharpoons Cu^{2+} + 4NH_3$$

The dissociation equilibrium constant or instability constant is represented as follows,

$$\alpha = \frac{\left[Cu^{2+}\right]\left[NH_3\right]^4}{\left[Cu(NH_3)_4\right]^{2+}} \qquad \text{---------(2)}$$

From (1) and (2) we can say that, the reciprocal of dissociation equilibrium constant (α) is called as formation equilibrium constant or stability constant (β).

$$\beta = \left(\frac{1}{\alpha}\right)$$

Significance of stability constants

The stability of coordination complex is measured in terms of its stability constant (β). Higher the value of stability constant for a complex ion, greater is the stability of the complex ion. Stability constant values of some important complexes are listed in table

Complex ion	Instability constant value (α)	stability constant value (β)
$\left[Fe(SCN)\right]^{2+}$	1.0×10^{-3}	1.0×10^{3}
$\left[Cu(NH_3)_4\right]^{2+}$	1.0×10^{-12}	1.0×10^{12}
$\left[Ag(CN)_2\right]^{-}$	1.8×10^{-19}	5.4×10^{18}
$\left[Co(NH_3)_6\right]^{3+}$	6.2×10^{-36}	1.6×10^{35}
$\left[Hg(CN)_4\right]^{2-}$	4.0×10^{-42}	2.5×10^{41}

By comparing stability constant values in the above table, we can say that among the five complexes listed, $\left[Hg(CN)_4\right]^{2-}$ is most stable complex ion and $\left[Fe(SCN)\right]^{2+}$ is least stable.

5.7.1. Stepwise formation constants and overall formation constants

When a free metal ion is in aqueous medium, it is surrounded by (coordinated with) water molecules. It is represented as $[MS_6]$. If ligands which are stronger than water are added to this metal salt solution, coordinated water molecules are replaced by strong ligands.

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